EXP 1 Study of various Network commands used in Linux and windows

Linux commands :

1. hostname - Displays the system hostname

```
abinash@Abiii:~$ hostname
Abiii
```

2. hostname -I - Displays the IP address of the system

```
abinash@Abiii:~$ hostname -I
172.28.250.180
```

3. ip addr - Shows network interface details and IP address

```
abinash@Abiii:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
```

4. ip route - Displays the routing table

```
abinash@Abiii:~$ ip route
default via 172.28.240.1 dev eth0 proto kernel
172.28.240.0/20 dev eth0 proto kernel scope link src 172.28.250.180
```

5. Ping google.com -c 4 - Tests connectivity to google

```
abinash@Abiii:~$ ping google.com -c 4
PING google.com (172.217.24.14) 56(84) bytes of data.
64 bytes from tsa01s07-in-f14.1e100.net (172.217.24.14): icmp_seq=1 ttl=114 time=60.4 ms
64 bytes from tsa01s07-in-f14.1e100.net (172.217.24.14): icmp_seq=2 ttl=114 time=106 ms
64 bytes from tsa01s07-in-f14.1e100.net (172.217.24.14): icmp_seq=3 ttl=114 time=278 ms
64 bytes from tsa01s07-in-f14.1e100.net (172.217.24.14): icmp_seq=4 ttl=114 time=66.8 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3148ms
rtt min/avg/max/mdev = 60.387/127.673/277.838/88.412 ms
```

6. Netstat -tuln - Lists active listening TCP/UDP Ports

```
abinash@Abiii:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 10.255.255.254:53       0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 10.255.255.254:53       0.0.0.0:*
udp        0      0 127.0.0.1:323           0.0.0.0:*
udp6       0      0 ::1:323                 :::*
```

7. SS - tuln - Displays Socket Statistics

```
abinash@Abiii:~$ ss -tuln
Netid       State       Recv-Q      Send-Q           Local Address:Port           Peer Address:Port      Process
udp         UNCONN      0           0                  127.0.0.54:53                0.0.0.0:*
udp         UNCONN      0           0               127.0.0.53%lo:53                0.0.0.0:*
udp         UNCONN      0           0              10.255.255.254:53                0.0.0.0:*
udp         UNCONN      0           0                  127.0.0.1:323                0.0.0.0:*
udp         UNCONN      0           0                      [::1]:323                   [::]:*
tcp         LISTEN      0           1000           10.255.255.254:53                0.0.0.0:*
tcp         LISTEN      0           4096            127.0.0.53%lo:53                0.0.0.0:*
tcp         LISTEN      0           4096               127.0.0.54:53                0.0.0.0:*
```

8. arp -a - Displays the ARP cache

```
abinash@Abiii:~$ arp -a
Abiii.mshome.net (172.28.240.1) at 00:15:5d:bf:c5:18 [ether] on eth0
```

Or

9. nslookup google.com - Finds the IP address of domain

Or

```
PS C:\Users\Abinash> nslookup google.com
Server:  UnKnown
Address:  10.179.89.77

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out
```

10. traceroute google.com - Shows the route taken to reach google

```
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    google.com
Addresses:  142.250.183.14
          2404:6800:4007:80a::200e
```

Windows Commands :-

1. hostname - Displays the computer name

```
PS C:\Users\Abinash> hostname
Abiii
```

2. ipconfig - Displays IP configuration

```
PS C:\Users\Abinash> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::37ec:f2b2:6e0b:ec11%14
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

3. ipconfig /all - Shows detailed network configuration

```
PS C:\Users\Abinash> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Abiii
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
```

4. Ping google.com - Test network connectivity

```
PS C:\Users\Abinash> ping google.com

Pinging google.com [2404:6800:4007:805::200e] with 32 bytes of data:
Reply from 2404:6800:4007:805::200e: time=80ms
Reply from 2404:6800:4007:805::200e: time=245ms
Reply from 2404:6800:4007:805::200e: time=115ms
Reply from 2404:6800:4007:805::200e: time=118ms

Ping statistics for 2404:6800:4007:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 80ms, Maximum = 245ms, Average = 139ms
```

5. tracert google.com - Displays the route to google

```
abinash@Abiii:~$ traceroute google.com
traceroute to google.com (142.250.67.46), 30 hops max, 60 byte packets
 1  Abiii.mshome.net (172.28.240.1)  0.576 ms  0.538 ms  0.512 ms
 2  10.179.89.77 (10.179.89.77)  5.734 ms  3.032 ms  3.007 ms
 3  * * *
 4  192.168.116.130 (192.168.116.130)  106.084 ms  105.982 ms  105.890 ms
 5  * * *
 6  * * *
 7  125.18.85.61 (125.18.85.61)  97.171 ms  93.302 ms 125.18.86.1 (125.18.86.1)  93.254 ms
 8  116.119.161.24 (116.119.161.24)  93.739 ms * 116.119.164.11 (116.119.164.11)  93.473 ms
 9  * * 142.250.169.206 (142.250.169.206)  102.246 ms
10  142.251.71.193 (142.251.71.193)  102.366 ms 142.251.70.211 (142.251.70.211)  102.065 ms 142.251.71.193 (142.251.71.1
93)  102.019 ms
11  142.251.55.219 (142.251.55.219)  101.316 ms  101.199 ms  101.152 ms
12  pnmaaa-bb-in-f14.1e100.net (142.250.67.46)  101.124 ms  102.772 ms  102.731 ms
```

6. arp -a - displays the ARP cache

```
PS C:\Users\Abinash> arp -a

Interface: 10.179.89.34 --- 0xa
  Internet Address      Physical Address      Type
  10.179.89.77          c6-cc-78-13-90-9e     dynamic
  10.179.89.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

7. Netstat -an - Lists active connections and

```
PS C:\Users\Abinash> netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:3306           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:33060          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
```

8. nslookup google.com - Resolves domain name to IP address

```
PS C:\Users\Abinash> tracert google.com

Tracing route to google.com [2404:6800:4007:805::200e]
over a maximum of 30 hops:

  1    13 ms     2 ms     1 ms  2401:4900:cadd:55a9::dc
  2   190 ms   102 ms   101 ms  2401:4900:1:a819::3
  3   117 ms   101 ms   101 ms  2401:4900:1:a88b::2
  4   422 ms   106 ms   211 ms  2401:4900:1:a88a::1
  5     *        *        *     Request timed out.
  6    89 ms    49 ms    92 ms  2404:a800:3a00:300::81
  7    62 ms   182 ms    81 ms  2404:a800::92
  8   207 ms   156 ms    65 ms  2001:4860:1:1::674
  9   129 ms   124 ms    91 ms  2001:4860:0:1::a03d
 10   365 ms   143 ms    60 ms  2001:4860:0:1::5649
 11    56 ms   158 ms   110 ms  pnmaaa-bb-in-x0e.1e100.net [2404:6800:4007:805::200e]

Trace complete.
```

9. getmac - Displays MAC address of System

```
PS C:\Users\Abinash> getmac

Physical Address    Transport Name
=================== ==========================================================
9C-C7-D3-2B-93-2E   \Device\Tcpip_{49FAE740-B542-40D6-8512-A5D8BB1C3AB6}
0A-00-27-00-00-0E   \Device\Tcpip_{774623B6-08A7-40BE-9583-94C3A1F03C5A}
```

10. route print - Displays the routing table

```
PS C:\Users\Abinash> route print
===========================================================================
Interface List
 14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
 11...92 c7 d3 2b 93 2e ......Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #2
  8...9e c7 d3 2b 93 2e ......Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC #3
 10...9c c7 d3 2b 93 2e ......Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
  1...........................Software Loopback Interface 1
 49...00 15 5d bf c5 18 ......Hyper-V Virtual Ethernet Adapter
===========================================================================
```

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